Vector Basics

```
vector<int> v;
v.push_back(23);
v.push_back(42);
v.pop_back();
```

No parameter.
Always removes last element

v = [23] ~~[42]~~
v[0] ~~v[1]~~

Other features: v.size() gives # elements.

v.front() ≡ v[0]

v.back() ≡ v[v.size()-1] // last element

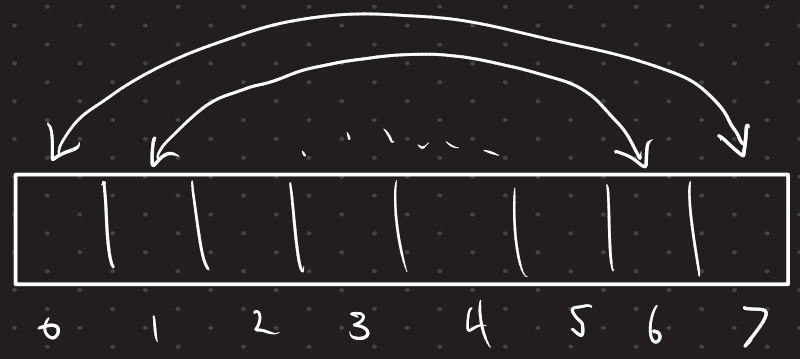
v.clear(): remove <u>all</u> elements

Warm up exercise: write a function that will reverse the order of the elements of a vector (of int).

E.g. if v = [1,2,3], after calling reverse(v), v = [3,2,1].

Prototype: void reverse(vector<int> & V);

Sketch:



(indexes)

V[0] ⟷ V[7]

V[1] ⟷ V[6]

⋮

V[i] ⟷ V[n-1-i]

say n = V.size()

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vectors as function parameters: best practice:

⊗ default to const + reference, e.g.

f(const vector<int> & V);

v is not changable by f

original is not copied.

⊗ The & improves performance.

The const simplifies things (rules out side effects)